

Multiplication and Inverse Matrices

Five ways to see a matrix product, all giving the same numbers. Then the inverse: when it exists, and how Gauss–Jordan finds it.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. The entry rule and shapes

Each entry of AB is a row of A times a column of B . Which shapes can be multiplied.

2. By columns and by rows

Columns of AB mix the columns of A ; rows of AB mix the rows of B . Many portfolios at once.

3. Columns times rows

AB as a sum of simple pieces, one per fund.

4. Block multiplication

Cut matrices into blocks and multiply the blocks.

5. The inverse, and when it exists

$A^{-1}A = I$. No inverse when some trade changes nothing: $Ax = 0$ with $x \neq 0$.

6. Gauss–Jordan: 2×2

Find A^{-1} by eliminating $[A \mid I]$ until it becomes $[I \mid A^{-1}]$.

7. Gauss–Jordan: 3×3 , and why it works

The three-fund portfolio, step by step. $E[A \mid I] = [I \mid E]$, so $E = A^{-1}$.

8. Using the inverse

Every target at once, building-block trades, the 2×2 formula, and when not to compute A^{-1} .

METHODS IN THIS LESSON

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PART 2 Prerequisites

Click any link to review it, then come back.

- Matrix $\times$ vector, two ways (Lesson 1): Ax mixes the columns of A .
- Singular matrices (Lesson 1): funds that add no new direction.
- Elimination on the augmented matrix (Lesson 2): the portfolio eliminated to U .
- Elimination matrices and "row $\times$ matrix mixes rows" (Lesson 2).
- Undoing one step (Lesson 2): the inverse of a single elimination matrix.

PART 3 Explanation

THE RUNNING EXAMPLE: MANY CLIENTS, ONE SET OF FUNDS

Our funds, as in every lesson:

Fund	Return per ₹1 lakh	Market exposure (beta) per ₹1 lakh
Nifty 50 index fund	₹12 thousand (12%)	1.0
Government bond fund	₹7 thousand (7%)	0
Midcap fund	₹15 thousand (15%)	1.2

Until now we multiplied the fund matrix A by **one** portfolio at a time: $Ax = b$. A desk has **many** clients. So we stack their portfolios side by side, one client per column, into a matrix X . With two funds and two clients:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget used per ₹1 lakh} \\ \leftarrow \text{return earned per ₹1 lakh} \end{array} \quad X = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} \begin{array}{l} \leftarrow \text{₹ lakh in Nifty} \\ \leftarrow \text{₹ lakh in bonds} \end{array}$$

Column 1 of X is client 1: ₹7 lakh Nifty, ₹3 lakh bonds. **Column 2** is client 2: ₹4 lakh Nifty, ₹6 lakh bonds. The product AX will give every client's (budget, return) at once. The **inverse** A^{-1} will then do the reverse: targets in, portfolios out.

1. The entry rule and shapes

Start from what we already know. A dot product of two lists of the same length multiplies matching entries and adds them: $(1, 1) \cdot (7, 3) = 1 \times 7 + 1 \times 3 = 10$. Lesson 1 used it to compute Ax row by row. A matrix product is nothing more than many dot products, one for every (row, column) pair.

METHOD · ENTRY RULE (ROW TIMES COLUMN)

The entry in row i and column j of $C = AB$ is the dot product of **row i of A** with **column j of B** :

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik} b_{kj}.$$

The index k walks along row i of A and, at the same time, down column j of B .

All four entries of AX , one at a time

1 WHAT WE DO

Find c_{11} , the entry in row 1, column 1.

HOW

Row 1 of A is $(1, 1)$. Column 1 of X is $(7, 3)$. Dot product: $1 \times 7 + 1 \times 3 = 7 + 3 = 10$.

MEANING

Row 1 of A is the budget rule, column 1 of X is client 1. So $c_{11} = 10$ is **client 1's budget**: ₹10 lakh.

2 WHAT WE DO

Find c_{12} (row 1, column 2).

HOW

Row 1 of A , $(1, 1)$, with column 2 of X , $(4, 6)$: $1 \times 4 + 1 \times 6 = 4 + 6 = 10$.

MEANING

Client 2's budget: also ₹10 lakh.

3 WHAT WE DO

Find c_{21} (row 2, column 1).

HOW

Row 2 of A , $(12, 7)$, with column 1 of X , $(7, 3)$: $12 \times 7 + 7 \times 3 = 84 + 21 = 105$.

MEANING

Client 1's return: ₹7 lakh in Nifty earns ₹84 thousand, ₹3 lakh in bonds earns ₹21 thousand, total ₹105 thousand.

4 WHAT WE DO

Find c_{22} (row 2, column 2).

HOW

Row 2 of A , $(12, 7)$, with column 2 of X , $(4, 6)$: $12 \times 4 + 7 \times 6 = 48 + 42 = 90$.

WHAT WE HAVE NOW

All four entries:

$$AX = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 1 \cdot 3 & 1 \cdot 4 + 1 \cdot 6 \\ 12 \cdot 7 + 7 \cdot 3 & 12 \cdot 4 + 7 \cdot 6 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix} \begin{array}{l} \leftarrow \text{budgets} \\ \leftarrow \text{returns} \end{array}$$

Reading the sum formula for c_{21} : $k = 1$ gives $a_{21}b_{11} = 12 \times 7$ (the Nifty term), $k = 2$ gives $a_{22}b_{21} = 7 \times 3$ (the bond term). One term per fund.

Which shapes can be multiplied

In every dot product, the entries of row i of A (one per fund) are paired with the entries of column j of B (one per fund). For the pairing to work, **the length of a row of A must equal the length of a column of B .**

RULE · SHAPES

$(m \times n)$ times $(n \times p)$ gives $(m \times p)$. The two inner numbers must match; the two outer numbers give the shape of the answer. Working out C takes $m \times n \times p$ multiplications (mp entries, n multiplications each).

A	B	Allowed?	Result	Portfolio reading
2×2	2×2	yes ($2 = 2$)	2×2	2 rules, 2 funds, 2 clients
3×3	3×5	yes ($3 = 3$)	3×5	3 rules, 3 funds, 5 clients
2×3	2×3	no ($3 \neq 2$)	none	3 funds' coefficients but only 2 holdings per client
1×3	3×1	yes	1×1	a single number: a dot product
3×1	1×3	yes	3×3	a whole matrix: a column times a row (topic 3)

The order matters: XA is a different matrix

Both X and A are 2×2 , so XA is also allowed. Entry by entry:

$$XA = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 7 \cdot 1 + 4 \cdot 12 & 7 \cdot 1 + 4 \cdot 7 \\ 3 \cdot 1 + 6 \cdot 12 & 3 \cdot 1 + 6 \cdot 7 \end{bmatrix} = \begin{bmatrix} 55 & 35 \\ 75 & 45 \end{bmatrix} \neq AX.$$

Not only are the numbers different, they mean nothing: $7 \times 1 + 4 \times 12$ mixes client 1's Nifty amount with client 2's Nifty amount times a return. **$AB \neq BA$ in general** (Lesson 2 already warned us).

IN ONE LINE

Each entry of AB = a row of A dotted with a column of B ; the inner sizes must match.

▼ Check yourself: with three funds and four clients, what are the shapes of A , X and AX , and how many multiplications does AX need?

A is 3×3 (3 rules $\times$ 3 funds), X is 3×4 (3 funds $\times$ 4 clients), AX is 3×4 . Multiplications: $3 \times 3 \times 4 = 36$ (12 entries, 3 each).

2. By columns and by rows

Computing entries one at a time is correct, but it hides the structure. Strang's next two views use whole columns and whole rows.

METHOD · COLUMN WAY

Column j of AB is A times column j of B . Since A times a vector is a combination of A 's columns (Lesson 1), **every column of AB is a combination of the columns of A .**

1 WHAT WE DO

Get column 1 of AX from client 1's portfolio (7, 3) alone.

WHY

Column 1 of AX uses only column 1 of X : client 2's numbers never enter it (look back at the entry formulas: c_{11} and c_{21} use only 7 and 3).

HOW

$$A \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 84 \end{bmatrix} + \begin{bmatrix} 3 \\ 21 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}.$$

MEANING

7 copies of the Nifty arrow plus 3 copies of the bond arrow: Lesson 1's column picture for client 1.

2 WHAT WE DO

Get column 2 from client 2's portfolio (4, 6).

HOW

$$A \begin{bmatrix} 4 \\ 6 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 6 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 4 \\ 48 \end{bmatrix} + \begin{bmatrix} 6 \\ 42 \end{bmatrix} = \begin{bmatrix} 10 \\ 90 \end{bmatrix}.$$

WHAT WE HAVE NOW

Put the two columns side by side: $AX = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$, the same as the entry rule.

METHOD · ROW WAY

Row i of AB is row i of A times B . Since a row times a matrix is a combination of the matrix's rows (Lesson 2), **every row of AB is a combination of the rows of B .**

The rows of X are "₹ in Nifty for each client", (7, 4), and "₹ in bonds for each client", (3, 6).

1 WHAT WE DO

Row 1 of AX : row 1 of A is $(1, 1)$, so take $1 \times (\text{row 1 of } X) + 1 \times (\text{row 2 of } X)$.

HOW

$$1(7, 4) + 1(3, 6) = (7 + 3, 4 + 6) = (10, 10).$$

MEANING

Every client's **budget**, in one row: Nifty holdings plus bond holdings.

2 WHAT WE DO

Row 2 of AX : row 2 of A is $(12, 7)$, so take $12 \times (\text{row 1 of } X) + 7 \times (\text{row 2 of } X)$.

HOW

$$12(7, 4) + 7(3, 6) = (84, 48) + (21, 42) = (84 + 21, 48 + 42) = (105, 90).$$

MEANING

Every client's **return**: 12 per lakh of Nifty plus 7 per lakh of bonds.

View	One piece of AX is...	Question it answers
Column way	a whole column = one client	What does each client end up with?
Row way	a whole row = one rule	What is the budget (or return) across all clients?

IN ONE LINE

Columns of AB mix the columns of A ; rows of AB mix the rows of B .

▼ **Check yourself: a third client holds ₹5 lakh in each fund. What new column appears in AX , by the column way?**

$$5(1, 12) + 5(1, 7) = (5, 60) + (5, 35) = (10, 95): ₹10 \text{ lakh used, ₹95 thousand earned.}$$

3. Columns times rows

A row times a column gives one number. Turn it around: a **column times a row** gives a whole matrix. Each entry is (an entry of the column) $\times$ (an entry of the row):

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 \times 1 & 2 \times 6 \\ 3 \times 1 & 3 \times 6 \\ 4 \times 1 & 4 \times 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix} \quad (\text{Strang's example})$$

- Column 1 is $1 \times (2, 3, 4)$ and column 2 is $6 \times (2, 3, 4)$: every column is a multiple of the column.
- Row 1 is $2 \times (1, 6)$, row 2 is $3 \times (1, 6)$, row 3 is $4 \times (1, 6)$: every row is a multiple of the row.

Strang calls this a "really minimal matrix": all its rows lie on one line, and so do all its columns. Later lessons call it a **rank-one** matrix.

METHOD · COLUMN-TIMES-ROW (SUM OF OUTER PRODUCTS)

$AB = (\text{column 1 of } A)(\text{row 1 of } B) + (\text{column 2 of } A)(\text{row 2 of } B) + \cdots + (\text{column } n \text{ of } A)(\text{row } n \text{ of } B)$. Each term, a column times a row, is called an **outer product**.

1 WHAT WE DO

Multiply column 1 of A , $(1, 12)$ (the Nifty fund), by row 1 of X , $(7, 4)$ (each client's Nifty holding).

HOW

$$\begin{bmatrix} 1 \\ 12 \end{bmatrix} \begin{bmatrix} 7 & 4 \end{bmatrix} = \begin{bmatrix} 1 \times 7 & 1 \times 4 \\ 12 \times 7 & 12 \times 4 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 84 & 48 \end{bmatrix}.$$

MEANING

What the **Nifty holdings** contribute: ₹7 lakh and ₹4 lakh of budget, ₹84 and ₹48 thousand of return.

2 WHAT WE DO

Multiply column 2 of A , $(1, 7)$ (bonds), by row 2 of X , $(3, 6)$ (each client's bond holding).

HOW

$$\begin{bmatrix} 1 \\ 7 \end{bmatrix} \begin{bmatrix} 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \times 3 & 1 \times 6 \\ 7 \times 3 & 7 \times 6 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 21 & 42 \end{bmatrix}.$$

MEANING

What the **bond holdings** contribute.

3 WHAT WE DO

Add the two pieces, entry by entry.

HOW

$$\begin{bmatrix} 7 + 3 & 4 + 6 \\ 84 + 21 & 48 + 42 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}.$$

WHAT WE HAVE NOW

AX again, now split by fund.

Why it works: entry $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j}$ has one term per fund. The first outer product puts exactly the term $a_{i1}b_{1j}$ into every position (i, j) , and the second puts exactly $a_{i2}b_{2j}$. Adding them rebuilds every entry.

IN MARKETS: ATTRIBUTION BY SOURCE

"How much of each client's return came from Nifty and how much from bonds?" is called **return attribution**. Here it is literally the column-times-row view: Nifty gave client 1 ₹84 of their ₹105 thousand, bonds gave ₹21.

IN ONE LINE

AB is a sum of column $\times$ row pieces, one per inner index (here, one per fund).

4. Block multiplication

Matrices can be cut into smaller rectangles called blocks, and multiplied block by block.

METHOD · BLOCK MULTIPLICATION

Cut A and B into blocks so that the **column cuts of A match the row cuts of B** . Then multiply the blocks exactly as if they were numbers: $\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$ has top-left block $A_1B_1 + A_2B_3$, and so on.

Use the full three-fund matrix and three clients (the targets from Lesson 4). Cut A 's columns after fund 2, and X 's rows after fund 2:

$$A = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{array} \right] = [A_{\text{left}} \mid A_{\text{right}}], \quad X = \left[\begin{array}{ccc} 5 & 4 & 6 \\ 3 & 5 & 6 \\ \hline 2 & 1 & 3 \end{array} \right] = \begin{bmatrix} X_{\text{top}} \\ X_{\text{bottom}} \end{bmatrix}$$

A_{left} (3×2) holds the Nifty and bond columns, A_{right} (3×1) the midcap column. X_{top} (2×3) holds every client's Nifty and bond amounts, X_{bottom} (1×3) every client's midcap amount. The cut after 2 columns of A matches the cut after 2 rows of X , so:

$$AX = A_{\text{left}}X_{\text{top}} + A_{\text{right}}X_{\text{bottom}}.$$

1 WHAT WE DO

HOW

Compute $A_{\text{left}}X_{\text{top}}$, the Nifty-and-bond part of every client.

Entry rule, row by row. Row 1 (1, 1): $(1 \cdot 5 + 1 \cdot 3, 1 \cdot 4 + 1 \cdot 5, 1 \cdot 6 + 1 \cdot 6) = (8, 9, 12)$. Row 2 (12, 7): $(60 + 21, 48 + 35, 72 + 42) = (81, 83, 114)$. Row 3 (1, 0): $(5 + 0, 4 + 0, 6 + 0) = (5, 4, 6)$.

WHAT WE HAVE NOW

$$A_{\text{left}}X_{\text{top}} = \begin{bmatrix} 8 & 9 & 12 \\ 81 & 83 & 114 \\ 5 & 4 & 6 \end{bmatrix}.$$

2 WHAT WE DO

Compute $A_{\text{right}} X_{\text{bottom}}$, the midcap part. It is a column times a row (topic 3).

HOW

$(1, 15, 1.2)^T$ times $(2, 1, 3)$: row 1 = $1 \times (2, 1, 3) = (2, 1, 3)$; row 2 = $15 \times (2, 1, 3) = (30, 15, 45)$; row 3 = $1.2 \times (2, 1, 3) = (2.4, 1.2, 3.6)$.

WHAT WE HAVE NOW

$$A_{\text{right}} X_{\text{bottom}} = \begin{bmatrix} 2 & 1 & 3 \\ 30 & 15 & 45 \\ 2.4 & 1.2 & 3.6 \end{bmatrix}.$$

3 WHAT WE DO

Add the two blocks.

HOW

Row 1: $(8 + 2, 9 + 1, 12 + 3) = (10, 10, 15)$. Row 2: $(81 + 30, 83 + 15, 114 + 45) = (111, 98, 159)$. Row 3: $(5 + 2.4, 4 + 1.2, 6 + 3.6) = (7.4, 5.2, 9.6)$.

WHAT WE HAVE NOW

The three clients' targets, one client per column:

$$AX = \begin{bmatrix} 10 & 10 & 15 \\ 111 & 98 & 159 \\ 7.4 & 5.2 & 9.6 \end{bmatrix} \begin{matrix} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{matrix}$$

Large matrices are multiplied in blocks so that each block fits in a computer's fast memory, and so blocks can be multiplied at the same time on different processors.

IN ONE LINE

If the cuts match, blocks multiply like numbers: top-left = $A_1 B_1 + A_2 B_3$.

▼ **Check yourself (MIT problem 3.1):** with $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 5 & 6 \end{bmatrix}$, compare $AB + AC$ with $A(B + C)$.

$AB = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 0 & 1 \cdot 0 + 2 \cdot 0 \\ 3 \cdot 1 + 4 \cdot 0 & 3 \cdot 0 + 4 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$. $AC = \begin{bmatrix} 0 + 10 & 0 + 12 \\ 0 + 20 & 0 + 24 \end{bmatrix} = \begin{bmatrix} 10 & 12 \\ 20 & 24 \end{bmatrix}$. Sum: $\begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. And $B + C = \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix}$, so $A(B + C) = \begin{bmatrix} 1 + 10 & 0 + 12 \\ 3 + 20 & 0 + 24 \end{bmatrix} = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. Equal: matrix multiplication is **distributive**, $A(B + C) = AB + AC$.

5. The inverse, and when it exists

With numbers: $5 \times \frac{1}{5} = 1$, so $\frac{1}{5}$ "undoes" multiplying by 5. Every number except 0 has such an inverse; 0 doesn't, because $0 \times \text{anything} = 0$, never 1.

For matrices, the role of 1 is played by the identity matrix I . Check that it changes nothing, using the entry rule on our A :

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 12 & 1 \cdot 1 + 0 \cdot 7 \\ 0 \cdot 1 + 1 \cdot 12 & 0 \cdot 1 + 1 \cdot 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = A.$$

DEFINITION: INVERSE

A square matrix A is **invertible** (also called **non-singular**) if there is a matrix A^{-1} with $A^{-1}A = I$ and $AA^{-1} = I$. Strang: for square matrices, if one of these holds the other holds too, so a left inverse is automatically a right inverse.

Why only square matrices: if A is $m \times n$, then $A^{-1}A = I$ needs A^{-1} to be $n \times m$, and $AA^{-1} = I$ is $m \times m$ while $A^{-1}A$ is $n \times n$. Both can be "the identity" in the same sense only when $m = n$. (Rectangular "one-sided" inverses come in Lesson 31.)

Strang: whether A has an inverse is "the most important question you can ask about the matrix". Start with one that doesn't.

A matrix with no inverse: two identical funds

Lesson 1's two Nifty funds, both 12%: $A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Look for a portfolio change x that changes nothing:

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot (-1) \\ 12 \cdot 1 + 12 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 1 - 1 \\ 12 - 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The trade $x = (1, -1)$: put ₹1 lakh **more** in fund A and ₹1 lakh **less** in fund B. Budget unchanged, return unchanged. Now Strang's argument that no inverse can exist:

1 SUPPOSE an inverse A^{-1} did exist.
DO Multiply both sides of $Ax = 0$ on the left by A^{-1} : $A^{-1}(Ax) = A^{-1}0$.

2 LEFT SIDE Move the parentheses (associative): $A^{-1}(Ax) = (A^{-1}A)x = Ix = x$.
RIGHT SIDE Any matrix times the zero vector is the zero vector: $A^{-1}0 = 0$.

3 SO $x = 0$. But $x = (1, -1)$ is **not** zero. The assumption was impossible: A has no inverse.

PLAIN WORDS

A sends both the trade $(1, -1)$ and "do nothing" $(0, 0)$ to the same result $(0, 0)$. No matrix can undo that, because it would have to send $(0, 0)$ back to two different places.

RULE · SINGULAR TEST: $AX = 0$ WITH $X \neq 0$

If some **non-zero** vector x has $Ax = 0$, then A has **no inverse** (A is singular). In portfolio terms: a trade that changes nothing means the portfolio for a target is never unique, so it can't be undone. Quick signs (from the recitation): a whole row or column of zeros, or two equal rows or columns. For example, if column j is all zeros then $x = (1 \text{ in position } j, 0 \text{ elsewhere})$ gives $Ax = 0$.

The same in 3×3 : the balanced fund (half Nifty, half bonds) has column $(1, 9.5, 0.5)$. Try the trade $x = (1, 1, -2)$: buy ₹1 lakh Nifty, buy ₹1 lakh bonds, sell ₹2 lakh balanced.

$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 + 1 - 2 \\ 12 + 7 - 19 \\ 1 + 0 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Budget, return and exposure all unchanged: this matrix has no inverse either.

Strang's column reason: every column of a product AX is a combination of A 's columns (topic 2). For the two-Nifty matrix both columns are $(1, 12)$, so every combination lies on the line through $(1, 12)$. To get $AX = I$, column 1 of I , $(1, 0)$, would have to lie on that line, and it doesn't. So no X can give I .

IN ONE LINE

$A^{-1}A = I$; if some non-zero trade x has $Ax = 0$, no inverse exists.

6. Gauss–Jordan: 2×2

Now a matrix that **does** have an inverse, the two-fund matrix $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$ (its columns point in different directions). Write the unknown inverse with letters:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} a & c \\ b & d \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_I$$

By the column way, this is **two separate systems with the same matrix A :**

- - - -

- Column 1: $A \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, that is $a + b = 1$ and $12a + 7b = 0$.
- Column 2: $A \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, that is $c + d = 0$ and $12c + 7d = 1$.

We could solve each by elimination separately. Jordan's idea: carry **both right-hand sides at once**, by augmenting A with the whole identity matrix.

METHOD · GAUSS-JORDAN ELIMINATION

Write $[A \mid I]$. **Gauss:** eliminate downward until the left half is upper triangular. **Jordan:** keep going upward, clearing the entries above each pivot. **Finally:** divide each row by its pivot. When the left half has become I , the right half is A^{-1} . Named after Carl Friedrich Gauss and Wilhelm Jordan (1842–1899), a German geodesist who popularised the upward step.

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 12 & 7 & 0 & 1 \end{array} \right]$$

1 WHAT WE DO

Gauss: clear the 12 below the first pivot 1.

WHY

To make the left half upper triangular, as in Lesson 2.

HOW

Multiplier $\ell_{21} = 12/1 = 12$. New row 2 = row 2 – 12 × row 1, entry by entry: $12 - 12 \times 1 = 0$; $7 - 12 \times 1 = -5$; $0 - 12 \times 1 = -12$; $1 - 12 \times 0 = 1$.

WHAT WE HAVE NOW

$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -5 & -12 & 1 \end{array} \right]$. Upper triangular on the left: Gauss alone would stop here.

2 WHAT WE DO

Jordan: clear the 1 **above** the second pivot –5.

WHY

To turn the left half into the identity, every entry above the diagonal must become 0 too.

HOW

Multiplier = $1/(-5) = -0.2$, so new row 1 = row 1 – (–0.2) × row 2 = row 1 + 0.2 × row 2: $1 + 0.2 \times 0 = 1$; $1 + 0.2 \times (-5) = 0$; $1 + 0.2 \times (-12) = 1 - 2.4 = -1.4$; $0 + 0.2 \times 1 = 0.2$.

WHAT WE HAVE NOW

$\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & -5 & -12 & 1 \end{array} \right]$. The left half is diagonal.

3 WHAT WE DO

Divide row 2 by its pivot –5.

HOW

$$0/(-5) = 0; -5/(-5) = 1; -12/(-5) = 2.4; 1/(-5) = -0.2.$$

(Row 1's pivot is already 1.)

WHAT WE HAVE NOW

$$\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & 1 & 2.4 & -0.2 \end{array} \right] = [I \mid A^{-1}].$$

$$A^{-1} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix}$$

Full check, both orders. $A^{-1}A$, entry by entry:

$$\begin{bmatrix} -1.4 \cdot 1 + 0.2 \cdot 12 & -1.4 \cdot 1 + 0.2 \cdot 7 \\ 2.4 \cdot 1 - 0.2 \cdot 12 & 2.4 \cdot 1 - 0.2 \cdot 7 \end{bmatrix} = \begin{bmatrix} -1.4 + 2.4 & -1.4 + 1.4 \\ 2.4 - 2.4 & 2.4 - 1.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$AA^{-1} = \begin{bmatrix} 1 \cdot (-1.4) + 1 \cdot 2.4 & 1 \cdot 0.2 + 1 \cdot (-0.2) \\ 12 \cdot (-1.4) + 7 \cdot 2.4 & 12 \cdot 0.2 + 7 \cdot (-0.2) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -16.8 + 16.8 & 2.4 - 1.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

What A^{-1} tells us

A and A^{-1} answer opposite questions:

Matrix	You give it...	It gives you...	Question it answers
A	a portfolio (x, y) : ₹ lakh in Nifty and bonds	its results (B, R) : budget and return	"What does this portfolio do?"
A^{-1}	the results you want : (B, R)	the portfolio (x, y) that delivers them	"What portfolio hits my target?"

So A^{-1} is a ready-made answer machine: $\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} B \\ R \end{bmatrix} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix} \begin{bmatrix} B \\ R \end{bmatrix}$. Its four numbers can be read two ways, by columns and by rows.

Reading A^{-1} by columns: what one extra unit of a target costs

By the column way, $A^{-1}(B, R) = B \times (\text{column 1}) + R \times (\text{column 2})$. So each column is **the change in the portfolio needed for one extra unit of one target, with the other target held fixed**. First, one fact does all the work:

THE KEY FACT

Every lakh you switch **from bonds to Nifty** keeps the budget the same and adds $12 - 7 = ₹5$ **thousand** of return. Every lakh switched the other way removes ₹5 thousand. (This 5 is the same 5 that sits in $ad - bc = -5$.)

1 COLUMN 2

WHY THESE NUMBERS

$(0.2, -0.2)$: the change for +₹1 thousand of return, same budget.

The budget must not move, so the only option is to switch money between the funds. Each lakh switched from bonds to Nifty adds ₹5 thousand, so ₹1 thousand needs $1/5 = 0.2$ lakh switched: Nifty $+0.2$, bonds -0.2 .

CHECK

Budget: $0.2 - 0.2 = 0$ ✓. Return: $12(0.2) + 7(-0.2) = 2.4 - 1.4 = 1$ ✓.

IN ₹

Move ₹20,000 from the bond fund to the Nifty fund, and the yearly return rises by ₹1,000.

2 COLUMN 1

WHY THESE NUMBERS

$(-1.4, 2.4)$: the change for +₹1 lakh of budget, same return.

Think in two moves. First add the new ₹1 lakh to bonds: budget $+1$, but return $+7$ thousand, which we don't want. Then cancel that ₹7 thousand by switching money from Nifty back to bonds: at ₹5 thousand per lakh that is $7/5 = 1.4$ lakh. Total: bonds $1 + 1.4 = 2.4$, Nifty -1.4 .

CHECK

Budget: $-1.4 + 2.4 = 1$ ✓. Return: $12(-1.4) + 7(2.4) = -16.8 + 16.8 = 0$ ✓.

IN PLAIN WORDS

Extra money with no extra return can only be absorbed by tilting towards the low-return fund: the new ₹1 lakh goes into bonds, and some Nifty is swapped into bonds too.

Building any target from the two columns

Because $x = B \times (\text{column 1}) + R \times (\text{column 2})$, every target is just "budget $\times$ column 1 + return $\times$ column 2":

1 TARGET

HOW

₹10 lakh, ₹105 thousand return (the running example).

$$10 \times (-1.4, 2.4) + 105 \times (0.2, -0.2) = (-14, 24) + (21, -21) = (-14 + 21, 24 - 21) = (7, 3).$$

ANSWER

₹7 lakh Nifty, ₹3 lakh bonds, the answer from Lessons 1 and 2.

2 TARGET

HOW

₹10 lakh at 10%, i.e. ₹100 thousand.

$$10 \times (-1.4, 2.4) + 100 \times (0.2, -0.2) = (-14, 24) + (20, -20) = (6, 4).$$

ANSWER

₹6 lakh Nifty, ₹4 lakh bonds. Check: $6 + 4 = 10 \checkmark$, $72 + 28 = 100 \checkmark$.

3 TARGET

₹10 lakh at 12%, i.e. ₹120 thousand.

HOW

$(-14, 24) + 120 \times (0.2, -0.2) = (-14 + 24, 24 - 24) = (10, 0)$.

ANSWER

Everything in Nifty, which makes sense: only Nifty earns 12%.

Reading A^{-1} by rows: one formula per fund

Each row of A^{-1} gives one fund's amount directly from the targets:

$$\text{Nifty} = -1.4B + 0.2R, \quad \text{Bonds} = 2.4B - 0.2R.$$

A client with ₹12 lakh who wants ₹120 thousand: $\text{Nifty} = -1.4(12) + 0.2(120) = -16.8 + 24 = 7.2$ and $\text{bonds} = 2.4(12) - 0.2(120) = 28.8 - 24 = 4.8$. Check: $7.2 + 4.8 = 12 \checkmark$, $12(7.2) + 7(4.8) = 86.4 + 33.6 = 120 \checkmark$.

IN ONE LINE

A^{-1} turns "what I want" into "what to hold". Its columns are the price of one extra unit of each target; its rows are one formula per fund.

IN ONE LINE

$[A \mid I] \rightarrow [I \mid A^{-1}]$: eliminate down (Gauss), up (Jordan), then divide by the pivots.

▼ Check yourself (Strang's example): invert $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ by Gauss–Jordan.

Gauss: $\ell_{21} = 2$; row 2 $- 2 \times$ row 1: $(2 - 2, 7 - 6 \mid 0 - 2, 1 - 0) = (0, 1 \mid -2, 1)$. Jordan: row 1 $- 3 \times$ row 2: $(1 - 0, 3 - 3 \mid 1 + 6, 0 - 3) = (1, 0 \mid 7, -3)$. Pivots are already 1. So $A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$. Check row 1 of $A^{-1}A$: $(7 - 6, 21 - 21) = (1, 0) \checkmark$.

7. Gauss–Jordan: 3×3 , and why it works

The three-fund portfolio. Start from $[A \mid I]$ (rows: budget, return, exposure):

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 12 & 7 & 15 & 0 & 1 & 0 \\ 1 & 0 & 1.2 & 0 & 0 & 1 \end{array} \right]$$

Gauss: downward

1 WHAT WE DO

Clear the 12 in row 2, column 1.

HOW

$\ell_{21} = 12/1 = 12$. Row 2 - 12 × row 1, all six entries: $12 - 12 = 0$; $7 - 12 = -5$; $15 - 12 = 3$; $0 - 12 = -12$; $1 - 0 = 1$; $0 - 0 = 0$.

WHAT WE HAVE NOW Row 2 = $(0, -5, 3 \mid -12, 1, 0)$.

2 WHAT WE DO

Clear the 1 in row 3, column 1.

HOW

$\ell_{31} = 1/1 = 1$. Row 3 - 1 × row 1: $1 - 1 = 0$; $0 - 1 = -1$; $1.2 - 1 = 0.2$; $0 - 1 = -1$; $0 - 0 = 0$; $1 - 0 = 1$.

WHAT WE HAVE NOW Row 3 = $(0, -1, 0.2 \mid -1, 0, 1)$.

3 WHAT WE DO

Clear the -1 in row 3, column 2, using the second pivot -5.

HOW

$\ell_{32} = (-1)/(-5) = 0.2$. Row 3 - 0.2 × row 2: $0 - 0 = 0$; $-1 - 0.2(-5) = -1 + 1 = 0$; $0.2 - 0.2(3) = 0.2 - 0.6 = -0.4$; $-1 - 0.2(-12) = -1 + 2.4 = 1.4$; $0 - 0.2(1) = -0.2$; $1 - 0.2(0) = 1$.

WHAT WE HAVE NOW Row 3 = $(0, 0, -0.4 \mid 1.4, -0.2, 1)$, and the left half is upper triangular:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -5 & 3 & -12 & 1 & 0 \\ 0 & 0 & -0.4 & 1.4 & -0.2 & 1 \end{array} \right]$$

Look at the right half: it is exactly $E = E_{32}E_{31}E_{21}$ from Lesson 2, the messy 1.4 included. The identity has quietly recorded every row operation.

Jordan: upward

1 WHAT WE DO

Clear the 3 above the third pivot (row 2, column 3).

HOW

Multiplier = $3/(-0.4) = -7.5$, so row 2 - (-7.5) × row 3 = row 2 + 7.5 × row 3: $0 + 0 = 0$; $-5 + 0 = -5$; $3 + 7.5(-0.4) = 3 - 3 = 0$

$$; -12 + 7.5(1.4) = -12 + 10.5 = -1.5; 1 + 7.5(-0.2) = 1 - 1.5 = -0.5; 0 + 7.5(1) = 7.5.$$

WHAT WE HAVE NOW Row 2 = $(0, -5, 0 \mid -1.5, -0.5, 7.5)$.

2 WHAT WE DO

HOW

Clear the 1 above the third pivot in row 1.

$$\text{Multiplier} = 1/(-0.4) = -2.5, \text{ so row } 1 + 2.5 \times \text{row } 3: 1 + 0 = 1; \\ 1 + 0 = 1; 1 + 2.5(-0.4) = 1 - 1 = 0; 1 + 2.5(1.4) = 1 + 3.5 = 4.5; 0 + 2.5(-0.2) = -0.5; 0 + 2.5(1) = 2.5.$$

WHAT WE HAVE NOW Row 1 = $(1, 1, 0 \mid 4.5, -0.5, 2.5)$.

3 WHAT WE DO

HOW

Clear the 1 above the second pivot -5 (row 1, column 2).

$$\text{Multiplier} = 1/(-5) = -0.2, \text{ so row } 1 + 0.2 \times \text{row } 2: 1 + 0 = 1; \\ 1 + 0.2(-5) = 0; 0 + 0 = 0; 4.5 + 0.2(-1.5) = 4.5 - 0.3 = 4.2; \\ -0.5 + 0.2(-0.5) = -0.5 - 0.1 = -0.6; 2.5 + 0.2(7.5) = 2.5 + 1.5 = 4.$$

WHAT WE HAVE NOW Row 1 = $(1, 0, 0 \mid 4.2, -0.6, 4)$. The left half is diagonal: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -0.4 \end{bmatrix}$.

Divide by the pivots

1 WHAT WE DO

HOW

Row 2 $\div (-5)$ and row 3 $\div (-0.4)$ (row 1's pivot is 1).

$$\text{Row 2: } -1.5/(-5) = 0.3; -0.5/(-5) = 0.1; 7.5/(-5) = -1.5. \\ \text{Row 3: } 1.4/(-0.4) = -3.5; -0.2/(-0.4) = 0.5; 1/(-0.4) = -2.5.$$

WHAT WE HAVE NOW $[I \mid A^{-1}]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4.2 & -0.6 & 4 \\ 0 & 1 & 0 & 0.3 & 0.1 & -1.5 \\ 0 & 0 & 1 & -3.5 & 0.5 & -2.5 \end{array} \right] \quad A^{-1} = \begin{bmatrix} 4.2 & -0.6 & 4 \\ 0.3 & 0.1 & -1.5 \\ -3.5 & 0.5 & -2.5 \end{bmatrix}$$

Full check: all nine entries of $A^{-1}A$

Each entry is (row of A^{-1}) $\cdot$ (column of A). The columns of A are $(1, 12, 1)$, $(1, 7, 0)$, $(1, 15, 1.2)$.

Row of A^{-1}	· column 1 (1, 12, 1)	· column 2 (1, 7, 0)	· column 3 (1, 15, 1.2)
(4.2, -0.6, 4)	$4.2 - 7.2 + 4 = 1$	$4.2 - 4.2 + 0 = 0$	$4.2 - 9 + 4.8 = 0$
(0.3, 0.1, -1.5)	$0.3 + 1.2 - 1.5 = 0$	$0.3 + 0.7 + 0 = 1$	$0.3 + 1.5 - 1.8 = 0$
(-3.5, 0.5, -2.5)	$-3.5 + 6 - 2.5 = 0$	$-3.5 + 3.5 + 0 = 0$	$-3.5 + 7.5 - 3 = 1$

1s on the diagonal, 0s elsewhere: $A^{-1}A = I$ ✓.

WHY GAUSS-JORDAN WORKS

Every row operation is a multiplication on the left by an elimination matrix (Lesson 2). Call the product of all of them E . Applying E to the long matrix acts on both halves at once (block multiplication, topic 4):

$$E[A \mid I] = [EA \mid EI] = [EA \mid E].$$

We stopped exactly when the left half became I , so $EA = I$. That says E undoes A : $E = A^{-1}$. And the right half is E itself. So the right half **has** to be A^{-1} . (Strang: "It tells us what E has to be.")

IN ONE LINE

The identity on the right records every row operation; when the left becomes I , the right is A^{-1} .

8. Using the inverse

Every client at once: $X = A^{-1}B$

The clients want targets B , and we need portfolios X with $AX = B$. Multiply both sides on the left by A^{-1} : $A^{-1}AX = A^{-1}B$, and since $A^{-1}A = I$, this is $X = A^{-1}B$. Each column of B (one client's target) gives one column of X (that client's portfolio). Row by row of A^{-1} :

Client (target)	Nifty = $(4.2, -0.6, 4) \cdot b$	Bonds = $(0.3, 0.1, -1.5) \cdot b$	Midcap = $(-3.5, 0.5, -2.5) \cdot b$
1: (10, 111, 7.4)	$42 - 66.6 + 29.6 = 5$	$3 + 11.1 - 11.1 = 3$	$-35 + 55.5 - 18.5 = 2$
2: (10, 98, 5.2)	$42 - 58.8 + 20.8 = 4$	$3 + 9.8 - 7.8 = 5$	$-35 + 49 - 13 = 1$
3: (15, 159, 9.6)	$63 - 95.4 + 38.4 = 6$	$4.5 + 15.9 - 14.4 = 6$	$-52.5 + 79.5 - 24 = 3$

So $X = \begin{bmatrix} 5 & 4 & 6 \\ 3 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix}$: client 1 holds (5, 3, 2), client 2 holds (4, 5, 1), client 3 holds (6, 6, 3), all in one product and matching Lessons 1–4.

The columns of A^{-1} are building-block trades

Read $AA^{-1} = I$ by the column way: A times column j of A^{-1} = column j of I . So column j of A^{-1} is the trade that changes rule j by exactly 1 and leaves every other rule unchanged. Each one checked in full:

Column of A^{-1}	Trade (Nifty, bonds, midcap, ₹ lakh)	Budget change	Return change	Exposure change
1	(4.2, 0.3, -3.5)	$4.2 + 0.3 - 3.5 = 1$	$50.4 + 2.1 - 52.5 = 0$	$4.2 + 0 - 4.2 = 0$
2	(-0.6, 0.1, 0.5)	$-0.6 + 0.1 + 0.5 = 0$	$-7.2 + 0.7 + 7.5 = 1$	$-0.6 + 0 + 0.6 = 0$
3	(4, -1.5, -2.5)	$4 - 1.5 - 2.5 = 0$	$48 - 10.5 - 37.5 = 0$	$4 + 0 - 3 = 1$

Use one: a client wants ₹5 thousand more return, nothing else changed. Take $5 \times$ column 2: $5 \times (-0.6, 0.1, 0.5) = (-3, 0.5, 2.5)$. That means move ₹3 lakh out of Nifty, put ₹0.5 lakh into bonds and ₹2.5 lakh into midcap. Check: budget $-3 + 0.5 + 2.5 = 0$; return $-36 + 3.5 + 37.5 = 5$; exposure $-3 + 0 + 3 = 0$ ✓. Desks call such columns **sensitivities**: how far each position must move per unit change in one target.

The 2 × 2 shortcut, and why it works

RULE · 2 × 2 INVERSE FORMULA

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, valid when $ad - bc \neq 0$: swap the diagonal, negate the off-diagonal, divide by $ad - bc$.

Why it works: multiply the bracket by the original matrix, entry by entry:

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix}.$$

Dividing by $ad - bc$ turns that into I . If $ad - bc = 0$ the division is impossible: no inverse. For the two funds, $a = 1, b = 1, c = 12, d = 7$: $ad - bc = 7 - 12 = -5$, and

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} 7 & -1 \\ -12 & 1 \end{bmatrix} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix},$$

the same as Gauss-Jordan. For the two identical Nifty funds, $ad - bc = 12 - 12 = 0$: no inverse, as we found. The number $ad - bc$ is the *determinant*, which gets whole lessons later.

Lesson 1's formula falls out: for budget B and return R , $x = A^{-1}(B, R)$ gives Nifty = $-1.4B + 0.2R = (R - 7B)/5$ and bonds = $2.4B - 0.2R = (12B - R)/5$, the formula derived by substitution in Lesson 1.

WATCH OUT: YOU RARELY NEED A^{-1} ITSELF

To solve $Ax = b$, don't compute A^{-1} and then multiply. Factor $A = LU$ once and do two triangular solves (Lesson 4): fewer operations and smaller rounding errors. Compute A^{-1} when you want its entries for their own sake, such as the sensitivities above.

IN ONE LINE

$X = A^{-1}B$ solves every target at once, and the columns of A^{-1} are the one-target-at-a-time trades.